

Design FMEA (qualitative version) — VBF-T1R2-DFMEA-01

Qualitative version, based on simulation signals only (annotated per protocol). Ratings: S (severity) and O (occurrence) $\in$ {High, Medium, Low}; basis = magnitude of the simulation value vs threshold. Simplified qualitative $S \times O$ matrix.

#	Failure mode	Failure cause	Simulation signal (detectability basis)	S	O	Design-side mitigation (implemented in V4b)
1	Lithium plating on negative electrode during fast charge	Local anode polarization at high current (kinetic + transport limitation), low temperature	`cell/r4_v4b_4c_df n.json:anode_potential_v` min +0.0215 V (margin thin — 21.5 mV above 0 V at 4C)	High	Low	N/P $\approx$ 1.35 (neg 96 μ m); negative particle 1.5 μ m (surface $\times$ 1.33); electrolyte bridge σ 1.8 S/m, D 6e-10, t 0.4 (estimate); negative porosity 0.32; cooling h = 60 keeps kinetics warm. Recommend: avoid 4C below 25 ° C; anode-reference-electrode monitoring in production
2	Cell temperature exceeding 60 ° C limit during fast charge	High-rate heat generation vs cooling capacity	4C T_max 326.46 K vs limit 333.15 K (margin 6.69 K)	High	Low	h = 60 W/m ² · K cold-plate-class cooling; thin collectors (8/6 μ m) and separator (9 μ m) shorten heat path; hA design-consistent in thermal-runaway coupling (0.3186 W/K)
3	Electrolyte oxidative decomposition under 4.7 V overcharge	4.7 V exceeds EC/EMC solvent stability window	Overcharge sim T_max 299.43 K, run-tr triggered = false (proxy caliber, lumped model); molecular HOMO / IE – EA vs 4.7 V not computed (`real_compute: false`, endorse skipped honestly)	Medium	Medium	Charge cut-off redundancy in BMS; follow-up true-DFT endorsement of electrolyte against 4.7 V recommended before production; additive screening (Stage 2 funnel) as next design round

#	Failure mode	Failure cause	Simulation signal (detectability basis)	S	O	Design-side mitigation (implemented in V4b)
4	Incomplete fast charge (voltage-limited CC stop)	Polarization at 4C reaches 4.2 V before full SOC	CC acceptance 4.53 Ah = 79% of 1C capacity	Low	High	Documented as performance characteristic; CV step / higher-t electrolyte / thinner electrodes as future levers (logged in evaluate R4)
5	Capacity shortfall	Under-loading / impedance growth	1C capacity 5.6997 Ah $\geq$ nominal 5.0 Ah	Medium	Low	Fine particles (2.0/1.5 μ m) for kinetics; porosity headroom (0.36/0.32) for transport
6	SEI growth / cycle aging	Calendar + cycling SEI accumulation	Not simulated in this case (aging protocol available; set is aging-capable, SEI kinetic rate constant 1e-12 m/s)	Low	Medium	Follow-up aging run recommended; SEI-suppressing coating is an available Stage 2 lever

Conclusion

Highest-risk items: (1) plating margin is thin (+21.5 mV) — mitigated by design but the dominant residual risk under cold/aged conditions; (2) 4.7 V oxidative stability unverified at molecular level — flagged for true-DFT endorsement. Mitigations above are already implemented in the nominated design. Complete FMEA (process/supplier failure modes) is N/A (beyond pure simulation boundary).